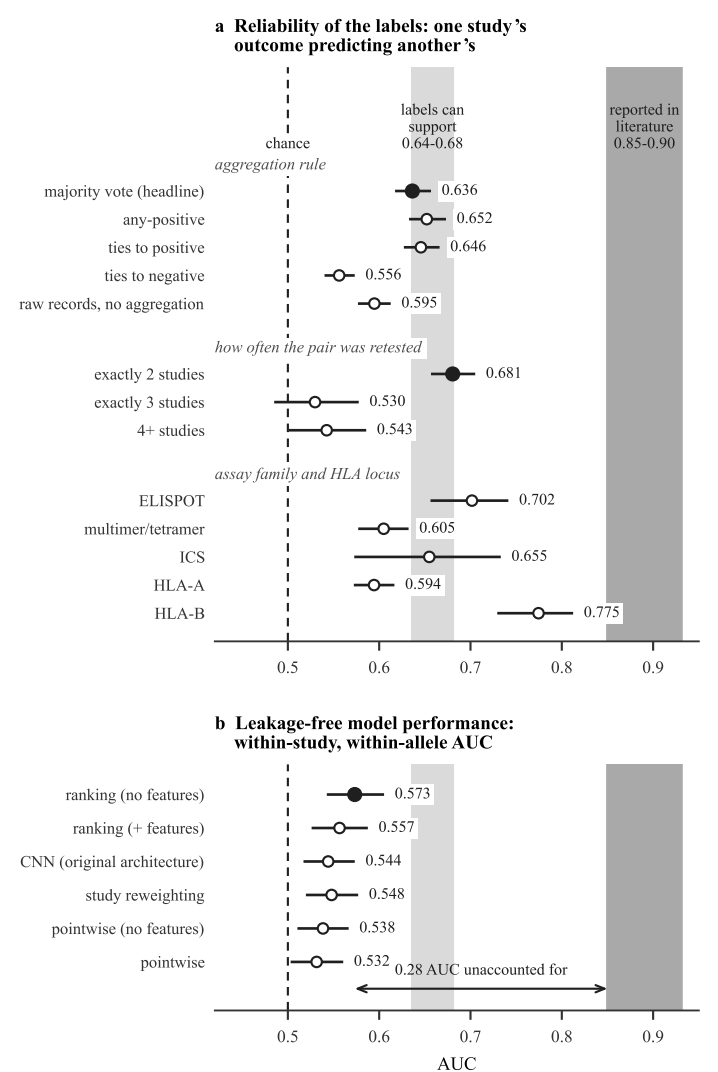

a Reliability of the labels: one study's outcome predicting another's
chance
labels can support 0.64-0.68
reported in literature 0.85-0.90
aggregation rule
majority vote (headline)
0.636
any-positive
0.652
ties to positive
0.646
ties to negative
0.556
raw records, no aggregation
0.595
how often the pair was retested
exactly 2 studies
0.681
exactly 3 studies
0.530
4+ studies
0.543
assay family and HLA locus
ELISPOT
0.702
multimer/tetramer
0.605
ICS
0.655
HLA-A
0.594
HLA-B
0.775
0.5
0.6
0.7
0.8
0.9
b Leakage-free model performance: within-study, within-allele AUC
ranking (no features)
0.573
ranking (+ features)
0.557
CNN (original architecture)
0.544
study reweighting
0.548
pointwise (no features)
0.538
pointwise
0.532
0.28 AUC unaccounted for
0.5
0.6
0.7
0.8
0.9
AUC